

Next, install the Python libraries required for the Generative AI Smart Speaker. These libraries support wake-word detection, voice capture, speech recognition, camera interfacing, AI model communication, and audio playback. Open the terminal in Raspberry Pi OS and install the required modules using:

```
sudo pip3 install <module_name>
```

For example:

```
sudo pip3 install openwakeword
```

Similarly, install the following libraries:

- **openwakeword** – Detects the wake word and activates the smart speaker
- **pyaudio** – Captures audio from the microphone
- **speech_recognition** – Converts speech into text
- **openai** – Provides access to generative AI and LLM services
- **numpy** – Performs numerical and data-processing operations
- **gtts** – Converts text responses into speech
- **pygame** – Plays audio responses through the speaker
- **opencv-python** – Captures and

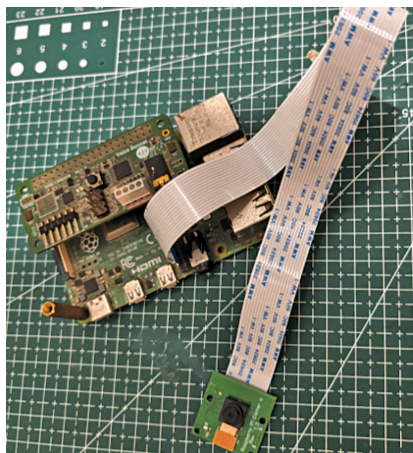


Fig. 6: Camera connection and voice bonnet mounted on the Raspberry Pi

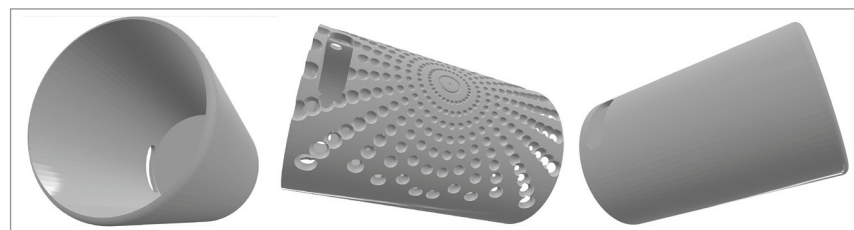


Fig. 7: Designed enclosure for the generative AI smart speaker

processes images from the camera

- **base64** – Encodes image data before sending it to AI models. This module is included with Python and does not require installation

After all the required modules have been installed, the Raspberry Pi is ready for software development and AI integration. Fig. 3 shows the installation of Python modules.

After installing all the required libraries, the next step is to prepare the software for the generative AI smart speaker. The code begins by importing the necessary Python modules and configuring the API key and the LLM model to be used. In this system, Gemini is used as the AI provider.

Next, define the assistant's name and the wake word used to activate the smart speaker. A custom wake word can be trained, or one of the pre-trained wake words available with OpenWakeWord can be used, such as Alexa, Jarvis, Mycroft, or Siri. In the prototype, the Alexa wake word was selected, and the assistant was also named Alexa for simplicity and ease of use. Fig. 4 shows the import of the Python module and the configuration of the AI name and wake word.

Next, create functions for commands used to control smart-home appliances or perform other automation tasks. These functions can operate Raspberry Pi GPIO pins or communicate with external devices and services through APIs. When a spoken command matches a predefined function, the corresponding action is executed immediately.

If the command does not match any predefined function, the speech

is sent to the configured LLM through its API. The AI model then generates an appropriate response, enabling the speaker to answer questions, tell jokes and stories, provide news and weather updates, and perform a wide range of conversational tasks. The system supports multiple LLM providers and can be switched easily to another model by updating the configuration and API key.

To extend its capabilities, additional functions for image analysis, translation, and transcription have been implemented. Using OpenCV and a connected camera, the smart speaker can capture images, identify objects, read text from books or documents, and describe its surroundings. It can also translate languages, transcribe speech, and assist with multilingual conversations. These features make the generative AI smart speaker significantly more capable than conventional smart speakers. Fig. 5 shows the code snippet used to configure the API connection to the generative AI model.

Connection

Fig. 6 shows the camera connection and the voice bonnet mounted on the Raspberry Pi. The voice bonnet plugs directly into the Raspberry Pi's GPIO header. It includes two screw terminals for speaker connections, allowing two speakers to be connected for stereo audio output. While a single speaker can also be used, it provides mono audio only. For a richer and more immersive smart-speaker experience, two speakers are recommended.

Next, connect the camera module to the Raspberry Pi's CSI camera port using the ribbon cable supplied with the camera. Once the camera and speakers are connected, the hardware setup is complete and ready for software testing.

Designed enclosure

Next, design a 3D-printed enclosure for the generative AI smart speaker. The enclosure can be customised to